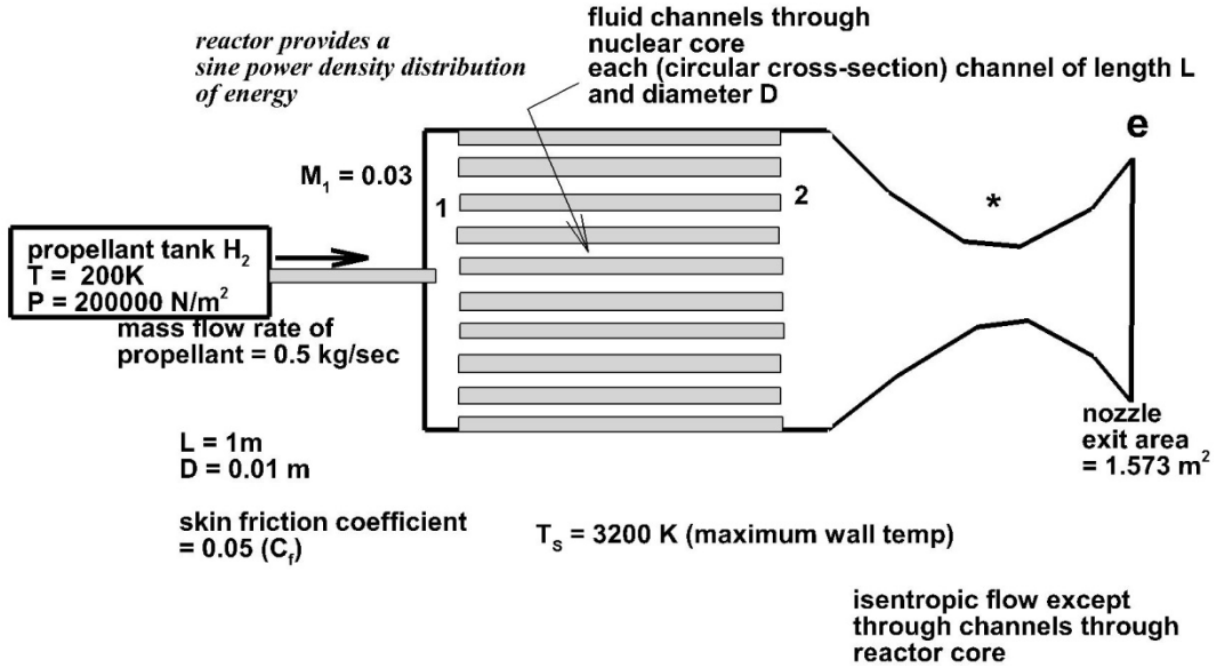
# Homework 6: Nuclear Propulsion

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**Answers:**

$x_{T_w=T_s} = 0.9031078\ m$

$\dot{Q} = 21.147\ MW$

$F =$

$I_{sp} =$

Conditions for sin power density distribution.

1) zero slope of $T_t$ at $x = 0$
2) max slope of $T_t$ at $x = L/2$
3) zero slope of $T_t$ at $x = L$

Additionally $T_s \rightarrow max(T_w)$ occurs at some $x$ between $x = L/2$ and $x = L$.

Can find the location where $T_w$ is equal to $T_s$ by using:

$$\left(\frac{x}{L}\right)_{T_w=T_s} = 1 + \frac{1}{\pi}\tan^{-1}\frac{-\pi D}{2C_f L} \rightarrow x_{T_w=T_s} = \boxed{0.9031078\ m} \tag{1}$$

Use this equation to find $\Delta T_m$:

$$T_s = T_{t_1} + \frac{2C_F L}{\pi D}\Delta T_m\left(1 + \sqrt{1 + \left(\frac{\pi D}{2C_f L}\right)^2}\right) \rightarrow \Delta T_m = 460.152\ K \tag{2}$$

We can then use this equation to find the total temperature at station 2:

$$T_{t_2} - T_{t_1} = \frac{4C_F L}{\pi D}\Delta T_m \rightarrow T_{t_2} = 3129.420\ K \tag{3}$$

Since we know the change in temperature through the reactor we can calculate the heat generated using:

$$\dot{Q} = \dot{m}c_p\left(T_{t_2} - T_{t_1}\right) = \boxed{21.147\ MW} \tag{4}$$

Where:

$$c_p = \frac{\gamma}{\gamma - 1}R = 14437.5\frac{J}{kgK} \tag{5}$$

The following equation, along with a root solver (scipy.optimize.fsolve), can be used to find the mach number at station 2:

$$\sqrt{\frac{T_{t_2}}{T_{t_1}}} = \frac{M_2}{M_1}\left[\frac{1 + \gamma M_1^2\left[1 - \frac{C_f L}{D}\right]}{1 + \gamma M_2^2\left[1 + \frac{C_f L}{D}\right]}\right]\left[\frac{1 + \frac{\gamma-1}{2}M_2^2}{1 + \frac{\gamma-1}{2}M_1^2}\right]^{\frac{1}{2}} \rightarrow M_2 = 0.138137 \tag{6}$$

The static pressure ratio is:

$$\frac{P_2}{P_1} = \left[\frac{1 + \gamma M_1^2\left[1 - \frac{C_f L}{D}\right]}{1 + \gamma M_2^2\left[1 + \frac{C_f L}{D}\right]}\right] = 0.85751166 \tag{7}$$

Then we can find the pressure at location 2:

$$\frac{P_{t_2}}{P_{t_1}} = \frac{P_2}{P_1}\left[\frac{1 + \frac{\gamma-1}{2}M_2^2}{1 + \frac{\gamma-1}{2}M_1^2}\right]^{\frac{\gamma}{\gamma-1}} = 0.86847317 \rightarrow P_{t_2} = 173.695\ kPa \tag{8}$$

This total pressure is also the total pressure at the exit of the nozzle and the chamber pressure.